

MLOps: Machine Learning Operations

Master in Business Analytics and Big Data (Part-Time) | Academic Year 2026–2027

COURSE DETAILS

Area	Computer Science & Artificial Intelligence
Number of sessions	12 (10 scheduled sessions + final examination)
Term / Category	3rd Term — Part-Time Electives / Elective Course
Language	English
Estimated total workload	40 hours
Professor	Juan Manuel Ramírez García
E-mail	jramirezg@faculty.ie.edu / juan.ramirez@es.ibm.com
Office Hours	On request, via e-mail booking
Affiliation	Data & AI Technical Sales Leader, IBM

PROFESSOR BIOGRAPHY

Juan Manuel Ramírez García is a Telecommunications Engineer from the Universidad Politécnica de Madrid (UPM) and holds an Executive MBA from IE Business School. He currently serves as Data & AI Technical Sales Leader for Spain, Portugal, Greece, and Israel at IBM, where for more than thirteen years he has led advanced analytics and Generative AI initiatives for large enterprise clients. His work sits at the intersection of business strategy and hands-on engineering: he has designed and delivered production machine learning and, more recently, Large Language Model (LLM) systems across Retail, Telecommunications, Banking, and Manufacturing. Before joining IBM he worked as a Predictive Analytics Specialist at SPSS, the statistical software firm acquired by IBM in 2010.

Because his day-to-day role involves helping organizations move models from prototype to dependable production service, Juan brings students the operational reality of AI — not only how models are built, but how they are deployed, monitored, governed, and kept trustworthy over time. This course is designed to be fully accessible to students expanding their careers into data and AI, including those without a prior technical or machine learning background: every concept is introduced from first principles and grounded in concrete industry examples.

SUBJECT DESCRIPTION

Organizations today aspire to be adaptive, client-centric, and efficient, and machine learning has moved from exploratory research to a core driver of business performance. Yet an industry-wide paradox persists: while data scientists are highly effective at training accurate models, the large majority of those models are never deployed to production, or they degrade quickly once they meet real-world data. Training a model is not the finish line; it is only the first mile of a continuous, high-stakes operational journey.

Machine Learning Operations (MLOps) is the engineering discipline that bridges the gap between data-science experimentation and reliable software production. It standardizes how AI assets are deployed, scaled, monitored, and governed so that they remain accurate, resilient, and audit-ready over time. As Tom Peters wrote in *Thriving on Chaos*, “we are drowning in information and starved for knowledge.” MLOps is what keeps that knowledge operational and dependable under changing, real-world conditions.

This elective decomposes the MLOps lifecycle into concrete, actionable steps. Students master the fundamentals of machine learning pipelines, Continuous Integration and Continuous Deployment (CI/CD), model and data monitoring, closed-loop retraining, and Explainable and Responsible AI. The course then extends these foundations to the operation of Generative AI: how monitoring, evaluation, and governance change when the deployed asset is a Large Language Model or a prompt-based application (LLMOps). Through industrial case studies and the analysis of real operational failures, students build the strategic and technical frameworks needed to lead resilient, enterprise-grade AI systems. No prior programming or machine learning experience is assumed; the course takes motivated professionals from foundational concepts to confident operational decision-making.

LEARNING OBJECTIVES

Upon successful completion of this elective, students will be able to:

- **Explain** the MLOps lifecycle end to end — from data ingestion and model development through deployment, monitoring, and retirement — and articulate why organizations fail to operationalize models.
- **Design** an end-to-end MLOps architecture connecting data pipelines, model versioning, automated deployment, and feedback loops for a defined business scenario.
- **Map** cross-functional MLOps roles (Data Scientist, ML Engineer, IT/DevOps, Product Owner, Risk & Compliance) to responsibilities and stage-gates within a realistic project roadmap.
- **Build** the logic of automated ML pipelines and CI/CD workflows, including versioning, automated testing of code and data, and safe rollout strategies (canary, blue-green, A/B).
- **Select and interpret monitoring metrics** for models in production, distinguishing data drift, concept drift, and performance decay, and defining alert thresholds and retraining triggers.
- **Extend monitoring and evaluation to Generative AI**, identifying the metrics and observability practices specific to LLM and prompt-based deployments (quality/faithfulness, hallucination and toxicity rates, latency, token cost, and user feedback).
- **Apply LLMOps concepts** to operate prompt-based and Retrieval-Augmented Generation (RAG) systems, including prompt versioning, evaluation pipelines, guardrails, and cost/latency management.
- **Integrate trust, compliance, and governance frameworks** — Explainable AI (XAI), Responsible AI, and regulatory standards such as the EU AI Act and the NIST AI Risk Management Framework — to mitigate bias, security, and governance risk.
- **Diagnose** pipeline failures and architectural bottlenecks through the critical evaluation of real enterprise deployment failure cases.

TEACHING METHODOLOGY

IE University's teaching method is collaborative, active, and applied. Students do not merely absorb lectures; they build, pressure-test, and debate modern MLOps systems, while the professor leads and facilitates that experiential process. The course alternates **Live Online sessions**, where all new concepts and topics are introduced, with **Discussion Forums** that act as a structured recap of the immediately preceding Live session: each forum poses three questions or small projects through which students consolidate and apply what was just presented. Two **self-paced asynchronous video lessons** deliver the technical deep-dives on monitoring metrics and governance. Every activity below is deliberately linked to the assessment scheme, so that what students practice is what they are evaluated on.

Learning Activity	Weighting	Estimated Student Dedication
Lectures (Live Online Sessions)	30.0 %	12.0 hours
Discussion Forums (Asynchronous Peer Collaboration)	15.0 %	6.0 hours
Asynchronous Video Lessons (Self-Paced, with Knowledge Checks)	10.0 %	4.0 hours
Exercises, Asynchronous Assignments & Field Work	15.0 %	6.0 hours
Group Project & Collaborative MLOps Planning	15.0 %	6.0 hours
Individual Studying & Self-Led Pipeline Diagnostics	15.0 %	6.0 hours
TOTAL	100.0 %	40.0 hours

Academic AI Policy: Responsible Adoption & Transparency

Generative AI (GenAI) tools are encouraged in this elective to support the conceptualization and prototyping of MLOps pipelines, scripting, and strategy formulation — this is, after all, a course about operating AI responsibly. However, GenAI must be used transparently and acknowledged rigorously. GenAI may **not** be used during the individual final exam or in direct-assessment quizzes. Any unattributed use of AI-generated content is treated as academic misconduct. To document usage, students should append the following to relevant submissions: *"I acknowledge the use of [AI*

PROGRAM

The course is organized into four interconnected modules that follow the full operational journey of a machine learning system — from foundations, to production pipelines, to monitoring and reliability, and finally to Generative AI operations and governance. **All new topics are introduced in the Live Online sessions.** Each **Discussion Forum** is a structured recap of the immediately preceding Live session, offering three questions or small projects that students develop to consolidate the concepts. The two **asynchronous video lessons** (Sessions 9 and 12) are self-paced technical deep-dives with knowledge checks.

Module 1/4 — Foundations of MLOps & Lifecycle Planning (Sessions 1-2)

Establishing the core paradigm, the organizational roles, and the end-to-end lifecycle.

SESSION 1 · LIVE ONLINEFoundations, Personas & the MLOps Lifecycle

Topics: Course objectives, mechanics, and assessment. Core definitions of MLOps; industry challenges; the "why" behind MLOps and the high failure rate of enterprise ML projects. The MLOps persona framework — roles, duties, and interfaces of Data Scientists, ML Engineers, IT/DevOps, and Risk & Compliance officers. Overview of the end-to-end lifecycle: data wrangling, model development, versioning, and baseline deployment.

Practical activity: MLOps business-value debate and diagnostic mapping of a failed deployment.

SESSION 2 · DISCUSSION FORUMRecap of Session 1 — Foundations, Personas & Lifecycle

Format: Asynchronous recap of the previous Live session. Develop the following three questions / small projects:

- Choose a real or hypothetical company and explain, in a short post, three reasons its ML models might never reach production. Which persona is best positioned to address each?
- Draft a one-page responsibility map (RACI-style) for a fictional recommendation-model project, assigning the Data Scientist, ML Engineer, IT/DevOps, Product Owner, and Risk/Compliance roles across the lifecycle stages.
- In your own words, define the difference between "training a model" and "operationalizing a model," and give one concrete business consequence of confusing the two.

Module 2/4 — Pipelines, Environments & Deployment (Sessions 3-6)

Building the automated engineering workflows that move models from local notebooks to reliable production.

SESSION 3 · LIVE ONLINEML Pipelines, Environments & Security

Topics: Translating exploratory notebooks into structured, reusable pipelines; ingestion, validation, and pre-processing automation. Multi-tier environment architecture (Dev, QA, Prod); auditability and data lineage; AI model security, vulnerabilities, and adversarial defense.

Practical activity: Constructing a model-risk mitigation blueprint.

SESSION 4 · DISCUSSION FORUMRecap of Session 3 — Pipelines, Environments & Security

Format: Asynchronous recap of the previous Live session. Develop the following three questions / small projects:

- Take the exploratory notebook provided on the virtual campus and sketch how you would restructure it into a reusable pipeline with separate ingestion, validation, and preprocessing steps. Submit the diagram and a short rationale.
- Explain why an organization needs separate Dev, QA, and Prod environments. What could go wrong if a model is promoted straight from a data scientist's laptop to production?
- Identify one adversarial or data-integrity threat to a deployed model and propose a mitigation. Which lineage or auditability control would help you detect it?

SESSION 5 · LIVE ONLINECI/CD & Deployment Strategies

Topics: Continuous Integration and Continuous Deployment for ML; automated testing for both code and data. Rollout strategies — canary releases, blue-green deployments, A/B testing — and serverless hosting models for zero-downtime deployment.

Practical activity: Selecting a rollout strategy for a zero-downtime banking model.

SESSION 6 · DISCUSSION FORUMRecap of Session 5 — CI/CD & Deployment

Format: Asynchronous recap of the previous Live session. Develop the following three questions / small projects:

- Design a CI/CD flow (diagram plus short description) for a fraud-detection model, showing which automated tests run on code and which run on data before deployment.
- Compare canary and blue-green deployments for a customer-facing bank model. Which would you recommend under a strict zero-downtime requirement, and why?
- Describe an A/B test you would run to decide whether a new model version outperforms the current one. Which metric and decision threshold would you use?

Module 3/4 — Monitoring, Reliability & Responsible AI (Sessions 7-9)

Ensuring long-term resilience through observability, retraining, and responsible automation.

SESSION 7 · LIVE ONLINEMonitoring, Explainability & Responsible AI

Topics: Continuous telemetry of production models; data drift, concept drift, and performance decay. Explainable AI (XAI) as an operational requirement. Closed-loop automation and automated retraining; bias mitigation, ethical guardrails, and compliance audits.

Practical activity: Configuring alert thresholds that trigger automated retraining.

SESSION 8 · DISCUSSION FORUMRecap of Session 7 — Monitoring, XAI & Responsible AI

Format: Asynchronous recap of the previous Live session. Develop the following three questions / small projects:

- For a churn-prediction model, define a monitoring plan: list the metrics you would track for data drift, concept drift, and performance decay, and the alert thresholds that would trigger retraining.
- Explain the difference between data drift and concept drift with a concrete example of each. How would your operational response differ between the two?
- Pick an algorithmic decision system (e.g., credit scoring). Identify one fairness risk and one explainability requirement, and describe how you would audit for each.

SESSION 9 · ASYNCHRONOUS VIDEO LESSONMonitoring Metrics in Depth: Classical ML and Prompt/LLM Deployments

Format: Self-paced video lesson with an embedded knowledge check (quiz).

Topics: A systematic catalogue of monitoring metrics across two families of deployment. For **classical ML models**: input/feature drift, prediction drift, concept drift, and performance metrics (accuracy, precision/recall, AUC, calibration) alongside operational metrics (latency, throughput, error rate). For **prompt-based and LLM deployments**: output-quality and faithfulness metrics, hallucination and toxicity rates, groundedness in retrieved context, response latency, token consumption and cost, and human/user feedback signals. How to choose metrics, set thresholds, and design dashboards and alerts for each case.

Knowledge check: Quiz mapping metrics to failure modes for both ML and LLM systems.

Module 4/4 — Generative AI Operations & Governance (Sessions 10-12)

Extending MLOps to Large Language Models and embedding governance across the lifecycle.

SESSION 10 · LIVE ONLINELLMOps & Enterprise Scenario Diagnostics

Topics: How operations change when the deployed asset is an LLM or prompt-based application: prompt versioning and management, Retrieval-Augmented Generation (RAG) pipelines, evaluation and "LLM-as-a-judge" pipelines, guardrails, caching, and cost/latency trade-offs. Contrasting the LLMOps lifecycle with classical MLOps. Intensive workshop analyzing real, high-cost operational AI failures (both classical ML and Generative AI), with root-cause analysis and mitigation, plus a synthesis of the course's four modules and emerging trends (edge deployment, agentic systems).

Practical activity: Designing the operational architecture for a RAG-based enterprise assistant and diagnosing an allocated failure case.

Preparation: Reading the allocated failure briefs posted on the virtual campus.

SESSION 11 · DISCUSSION FORUM

Recap of Session 10 — LLMOps & Diagnostics (Course Capstone)

Format: Asynchronous recap of the previous Live session. Develop the following three questions / small projects:

- Design the operational architecture for a RAG-based internal assistant. Show the components (retrieval, prompt management, evaluation, guardrails, monitoring) and note one cost or latency trade-off.
- Compare classical MLOps and LLMOps: give two concrete ways in which operating an LLM in production differs from operating a traditional ML model.
- Capstone synthesis:** take one enterprise failure case discussed in Session 10 and outline a corrective MLOps roadmap that spans the four course modules — foundations, pipelines and deployment, monitoring, and governance.

SESSION 12 · ASYNCHRONOUS VIDEO LESSON

AI Governance & Regulatory Readiness

Format: Self-paced video lesson with an embedded knowledge check (quiz).

Topics: Governance as an operational discipline. Regulatory and standards landscape, including the EU AI Act (risk tiers and obligations) and the NIST AI Risk Management Framework; documentation and audit-readiness (model cards, datasheets, decision logs); accountability, human oversight, and the governance of Generative AI specifically. How governance requirements translate into concrete monitoring, documentation, and control points across the MLOps pipeline.

Knowledge check: Quiz classifying AI use cases by regulatory risk tier and required controls.

Final Examination is administered at the end of the term (see Evaluation Method).

BIBLIOGRAPHY

Compulsory

- Treveil, M., & the Dataiku Team (2020). *Introducing MLOps: How to Scale Machine Learning in the Enterprise*. O'Reilly Media. ISBN 9781492083290.
- Huyen, C. (2022). *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. O'Reilly Media. ISBN 9781098107963.

Complementary / Recommended

- Gift, N., & Deza, A. (2021). *Practical MLOps: Operationalizing Machine Learning Models*. O'Reilly Media. ISBN 9781098103019.
- Huyen, C. (2025). *AI Engineering: Building Applications with Foundation Models*. O'Reilly Media. ISBN 9781098166304. (For the LLMOps and Generative AI operations sessions.)
- Google Cloud (Architecture Center). *MLOps: Continuous Delivery and Automation Pipelines in Machine Learning* (whitepaper, available online).
- European Union (2024). *Regulation (EU) 2024/1689 — Artificial Intelligence Act*. (For Session 12 — AI Governance.)
- National Institute of Standards and Technology — NIST (2023). *AI Risk Management Framework (AI RMF 1.0)*. (For Session 12 — AI Governance.)

Additional readings, case briefs, and technical notes are posted on the virtual campus for each module.

EVALUATION METHOD

Disclaimer: The assessment method may be subject to minor modifications. Should any change occur, students will be notified on the first day of class and provided with a written document detailing the modifications.

Your final grade combines individual knowledge, applied group work, continuous assessment, and active participation. Weights are transparent and each component is mapped to the learning objectives it evaluates, ensuring full alignment between objectives, activities, and assessment.

Assessment Component	Weight	Learning Objectives	Nature of Assessment
Individual Final Exam	35.0 %	LO 1, 2, 4, 5, 6, 8	Comprehensive individual exam on lifecycle logic, pipeline and CI/CD reasoning, monitoring/drift metrics, and governance frameworks.
Group Project (Workgroups)	35.0 %	LO 2, 3, 4, 5, 7, 9	Collaborative design and defense of an end-to-end MLOps roadmap for an assigned enterprise scenario.
Continuous Assessment — Quizzes & Video-Lesson Knowledge Checks	15.0 %	LO 1, 5, 6, 8	Short quizzes across modules, including the two knowledge checks embedded in the asynchronous video lessons (Sessions 9 and 12).
Class & Forum Participation	15.0 %	LO 1, 3, 6, 9	Continuous evaluation of live-session contributions and asynchronous discussion forums.
TOTAL	100.0 %	All Objectives	Fully grounded in applied practice.

A. Class & Forum Participation (15%)

Participation is measured on **quality, insight, and rigor — not on volume**. High-quality contributions connect MLOps concepts, empirical industry examples, and technological constraints, and students are expected to initiate and sustain debate in the asynchronous forums as well as in live sessions. Indicative rubric: *Excellent* — consistently advances the discussion with well-reasoned, evidence-based points; *Good* — regular, relevant contributions; *Satisfactory* — occasional participation of limited depth; *Insufficient* — minimal or off-topic participation. To keep dialogue valuable, students should avoid being *Repeaters* (restating points already made), *Ramblers* (long, low-impact tangents), or *the Distracted* (losing attention and later raising already-answered questions).

B. Group Project — Workgroups (35%)

Teams of 3-5 students act as a consulting MLOps team for an enterprise scenario that needs to move a complex AI prototype into reliable production. Deliverables include an MLOps roadmap, defined persona responsibilities, CI/CD pipeline schematics, a monitoring-and-retraining policy (covering both classical ML metrics and, where relevant, LLM/prompt metrics), and a governance framework. Indicative rubric dimensions (equally weighted): technical soundness of the architecture; depth and clarity of documentation; quality of monitoring and governance design; and the team's defense of its decisions. Full rubric and deliverable templates are posted on the virtual campus.

C. Continuous Assessment — Quizzes & Knowledge Checks (15%)

Short quizzes distributed across the modules assess ongoing understanding and provide timely feedback. This component explicitly includes the knowledge checks embedded in the two asynchronous video lessons (Session 9 — Monitoring Metrics; Session 12 — AI Governance), reinforcing the alignment between the self-paced content and the assessment.

D. Individual Final Exam (35%)

One final examination is administered at the end of the term, assessing individual understanding of the theoretical and technical frameworks covered. Students may bring **one single-sided A4 sheet** of handwritten formulas, architecture diagrams, or alert-threshold logic. No phones, smartwatches, tablets, or internet-connected devices are permitted. **To pass the course, students must obtain a minimum of 3.5 out of 10.0 in the final exam.** A score below this threshold results in a failing grade for the course, regardless of the weighted average.

Failing Grade and Reassessment Policy

In line with IE University academic regulations, a student who receives an overall failing grade has a single reassessment opportunity, scheduled between 5 and 10 working days after the official grade-review session. The maximum grade achievable in reassessment is a "Low Pass" (5.0); failure results in a definitive "Fail." Both the initial and reassessment grades are recorded on the transcript, but only the reassessment grade is used in the GPA. Students who fail the reassessment are academically restricted from continuing in the master's program.

UNIVERSITY POLICIES & CONDUCT CODES

Academic Integrity — Please consult the University's Academic Integrity Policy on the virtual campus. The Program Director may provide further indications.
Attendance — Please consult the University's Attendance Policy on the official portal. The Program Director may provide further indications.
Student Ethics — Please consult the University's Student Code of Conduct. The Program Director may provide further indications.